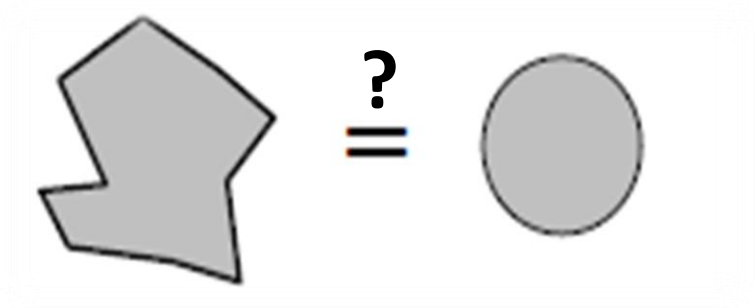
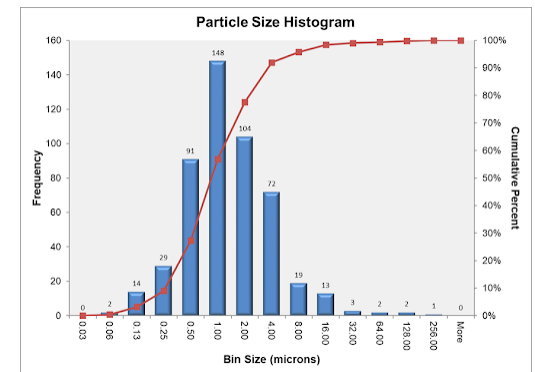
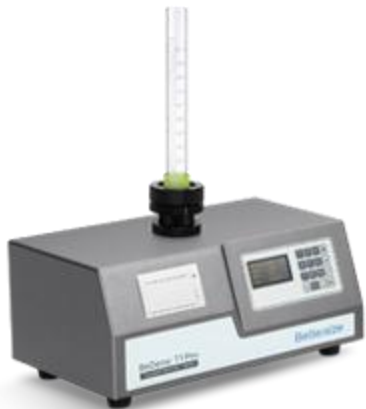




Micromeritics



Chapter Outline

- Introduction
- Particle size and size distribution
- Methods for determining particle size
- Particle shape and surface area
- Methods for determining particle surface area
- Derived Properties of powders

Objectives

-
- ❑ After the end of this topic ,students will be able to :
 - ❑ Know the fundamental and derived properties of individual and collection of particles
 - ❑ Describe the various methods used for determination of:
 - ❑ particle size and size distribution
 - ❑ Particle shape and surface area
 - ❑ densities, porosity, packing arrangement, flowability

Introduction

Micromeritics?

Is the science and technology of **small particles** that deals with the **fundamental** and **derived properties** of individual and collection of particles

Introduction...

► **Fundamental properties**

- These are properties from which other properties can be derived
 - Particle size and size distribution
 - Particle shape and surface area
 - Particle number and weight
 - Particle volume

Introduction...

► Derived properties

- Porosity
- Density
- Bulkiness or specific bulk volume

Application in Pharmacy

Why we study particle properties?

- **Physical properties of powder** are dependent on **particle size and size distribution**
 - ▶ Bulk density, compressibility, Porosity
- **Flow properties** of the powder
 - ▶ Spherical particles:- good flow property
 - ▶ Asymmetrical particles:- poor flow property

Application in Pharmacy...

- **Dose uniformity**

- ▶ Good flow properties of granules & powders are important in the manufacturing of tablets & capsules
- ▶ The flow of granules should be uniform in order to ensure precise weight of the tablet/capsule and drug content

Application in Pharmacy...

- **Release and dissolution**

- ▶ **Higher surface area** allows intimate contact of the drug with the dissolution fluids *in vivo* & increases the drug solubility & dissolution

- **Absorption and drug action**

- ▶ Higher the dissolution, faster the absorption & hence quicker & greater the drug action

Application in Pharmacy...

○ Physical stability

▶ Suspensions & emulsions

- Smaller the size of the particle, better the physical stability of the dosage form

Fundamental Properties ... *Particle size*

- ▶ Particles are 3-dimensional objects, and unless they are **perfect spheres**, they cannot be fully described by a single dimension such as a **radius or diameter**.
- ▶ Particles can be
 - **Symmetric**
 - **Asymmetric**

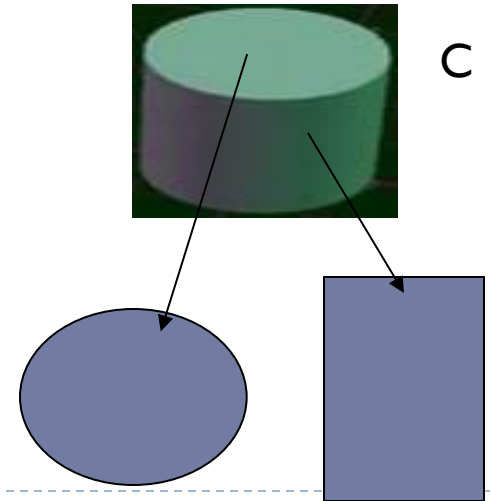
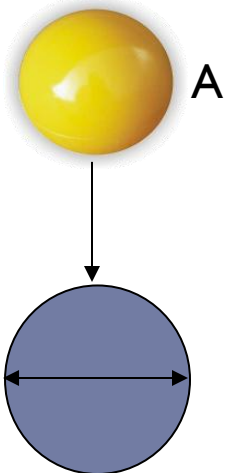
Fundamental Properties ... *Particle size*

Symmetric particles

- ▶ The size of particles of simple geometry (e. g. spheres and cubes) can be accurately and unambiguously determined using diameter and edge length.
- ▶ So, for a perfect sphere;

▶ **Surface area,** $S_A = \pi d^2$

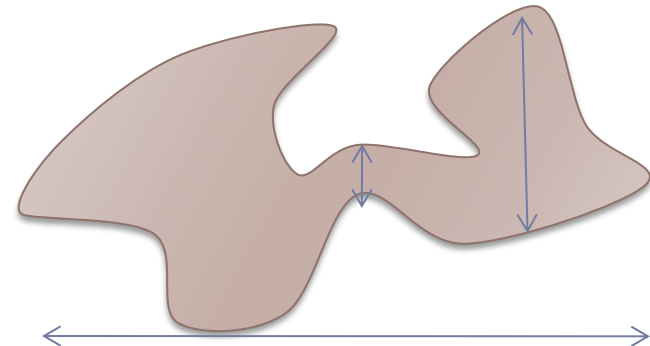
Volume, $V = \frac{\pi d^3}{6}$



Fundamental Properties ... *Particle size*

Asymmetric particles

- have a definite **surface area** and **volume** but being asymmetric, its apparent length varies with its orientation

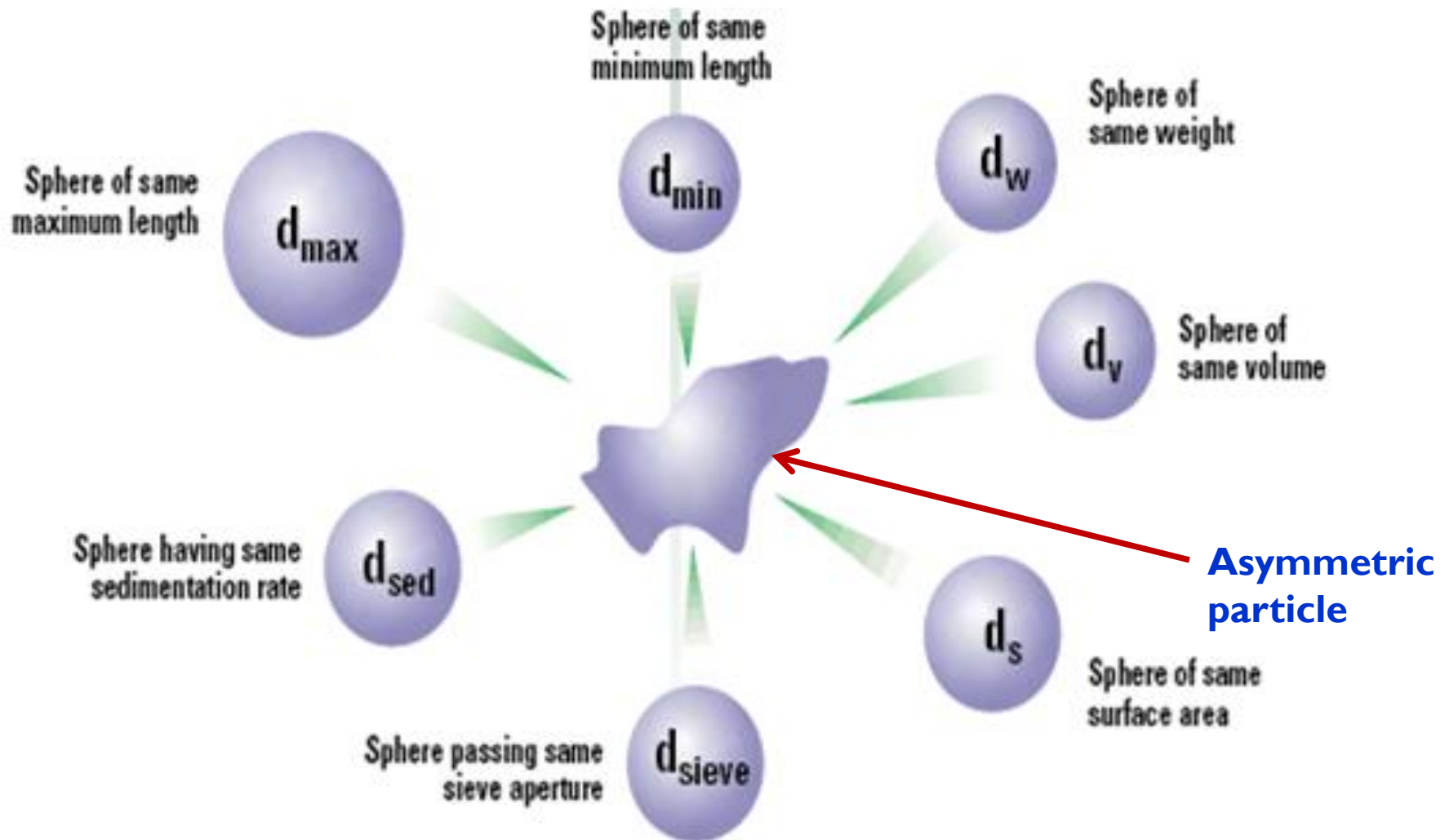


- Hence, it is difficult to express its size in terms of diameter

Fundamental Properties ... *Particle size*

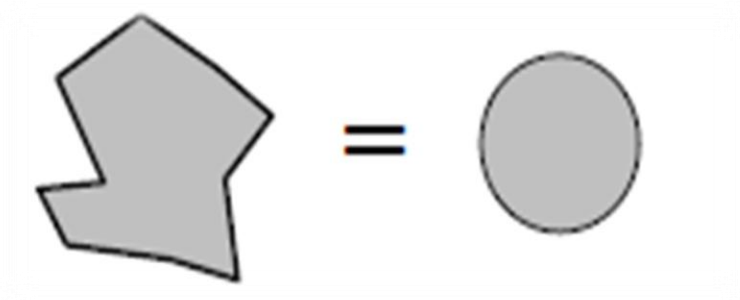
- ▶ Particle size instruments specify the size of irregularly shaped particles **using an equivalent spherical diameter**
- ▶ In this case, the particle size is defined by the diameter of an ***equivalent sphere*** having the **same property** as the actual particle

Fundamental Properties ... *Particle size*



Fundamental Properties ... *Particle size*

- ▶ Surface diameter (d_s): the diameter of a sphere having the **same surface area** as that of the asymmetric particle in question

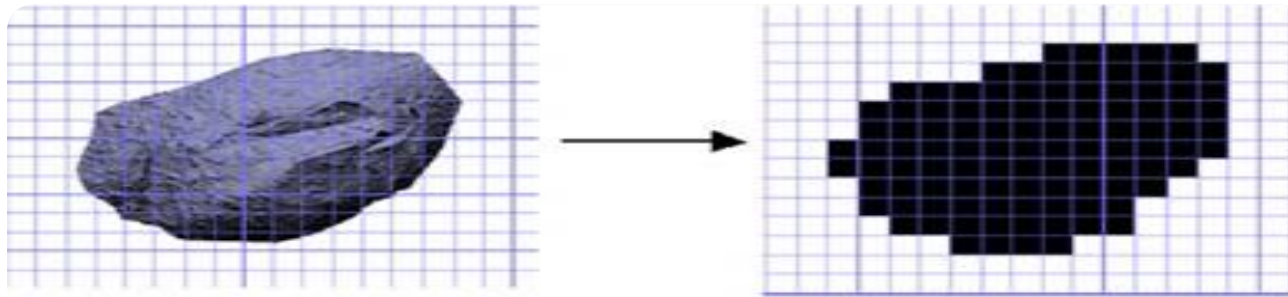


Fundamental Properties ... *Particle size*

- ▶ Stoke's diameter (d_{st}): the diameter of a sphere with **the same density** as the asymmetric particles in question and which undergoes sedimentation at **the same rate** as the asymmetric particles in a given fluid
 - d_{st} is usually determined using sedimentation methods

Fundamental Properties ... *Particle size*

- ▶ **Projected diameter** (d_p): the diameter of a sphere having the **same observed area** as that of the asymmetric particles in question when viewed **normal to its most stable plane**
 - Usually determined using **microscopic techniques**



- ▶ **Volume diameter** (d_v): the diameter of a sphere having the **same volume** as that of the asymmetric particles in question.

NB:- Equivalent spherical diameters for irregularly shaped particles determined in different ways are not numerically equal.

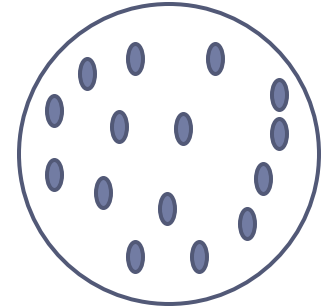
Each technique is not wrong – they are all right – it is just that a different property of the particle is being measured.

Fundamental Properties ... *Particle size distribution*

A powder may contain:

1. **Mono-disperse** particle

— Its characteristics can be described by a single diameter or equivalent diameter



2. **Poly-disperse** particle

— commonly encountered in pharmaceutical powders
— mixture of particles with **varying size and shape**



○ Cannot be described **by a single diameter**

Fundamental Properties ... *Particle size distribution*

- It is therefore necessary to know not only the size of a certain particle, but also **how many** particles of the same size exist in the sample
- Thus, we need an estimate of the **size range** present and the **number** or **weight** fraction of each particle size
- This is called the **particle size distribution** and from this, the **average particle size** of the collection of particles can be derived.

Fundamental Properties ... *Particle size distribution*

Size range (μm)	Mean size range (μm), d	Number of particles in each size range, n	nd	nd^2	nd^3
0.5-1.0	0.75	2			
1-1.5	1.25	10			
1.5-2.0	1.75	22			
2.0-2.5	2.25	54			
2.5-3.0	2.75	17			
3.0-3.5	3.25	8			
3.5-4.0	3.75	5			

Fundamental Properties ... *Particle size distribution*

- ▶ Two sample of powder may have the **same average diameter** but may not have the same frequency distribution.
- ▶ So, expression of the size in terms of average diameter may not give a clear expression of the **particle size distribution**

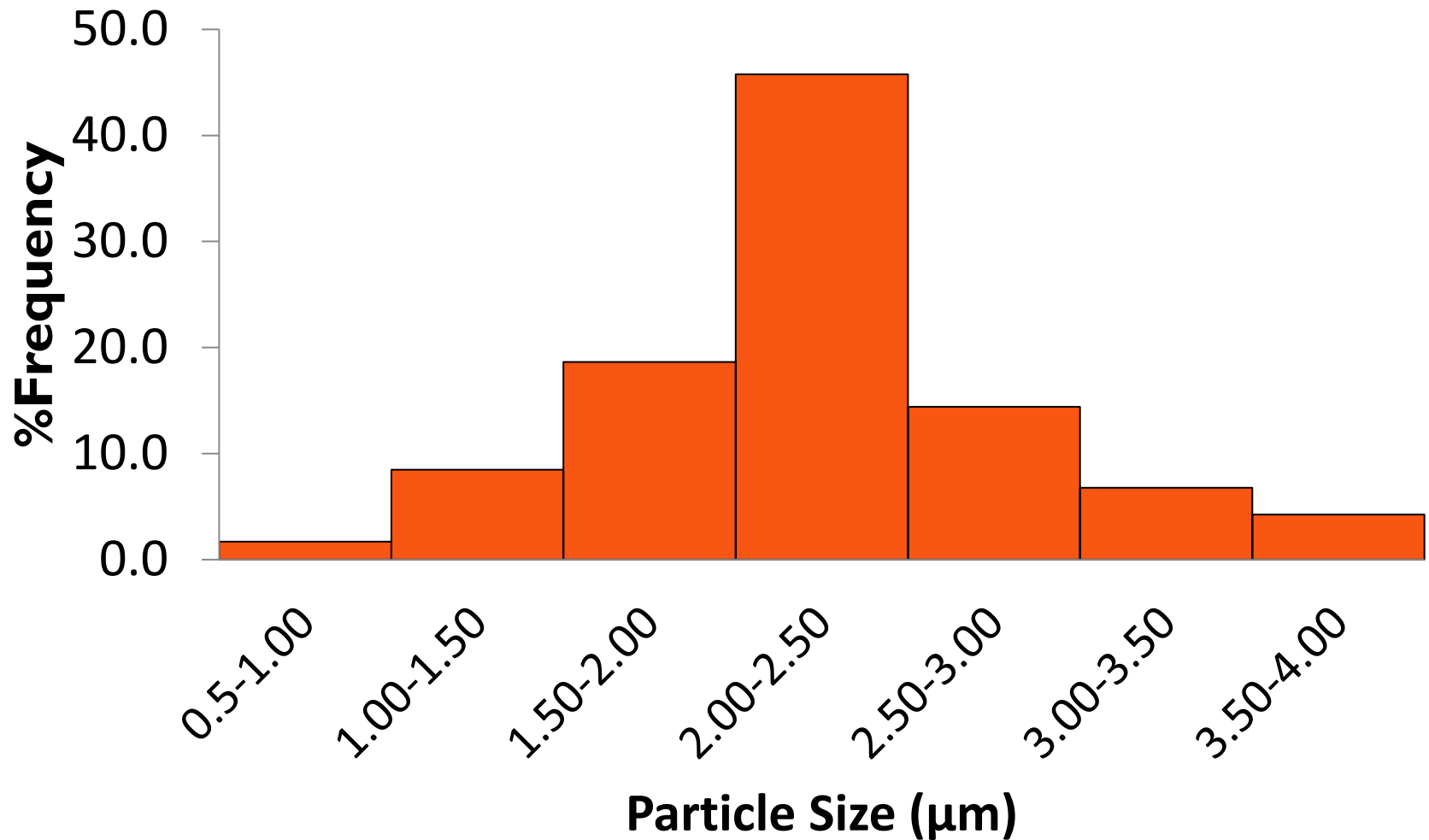
Fundamental Properties ... *Particle size distribution*

- ▶ It is common practice to represent this distribution in the form of either
 - Frequency distribution curve
 - Cumulative (undersize, oversize) distribution curve

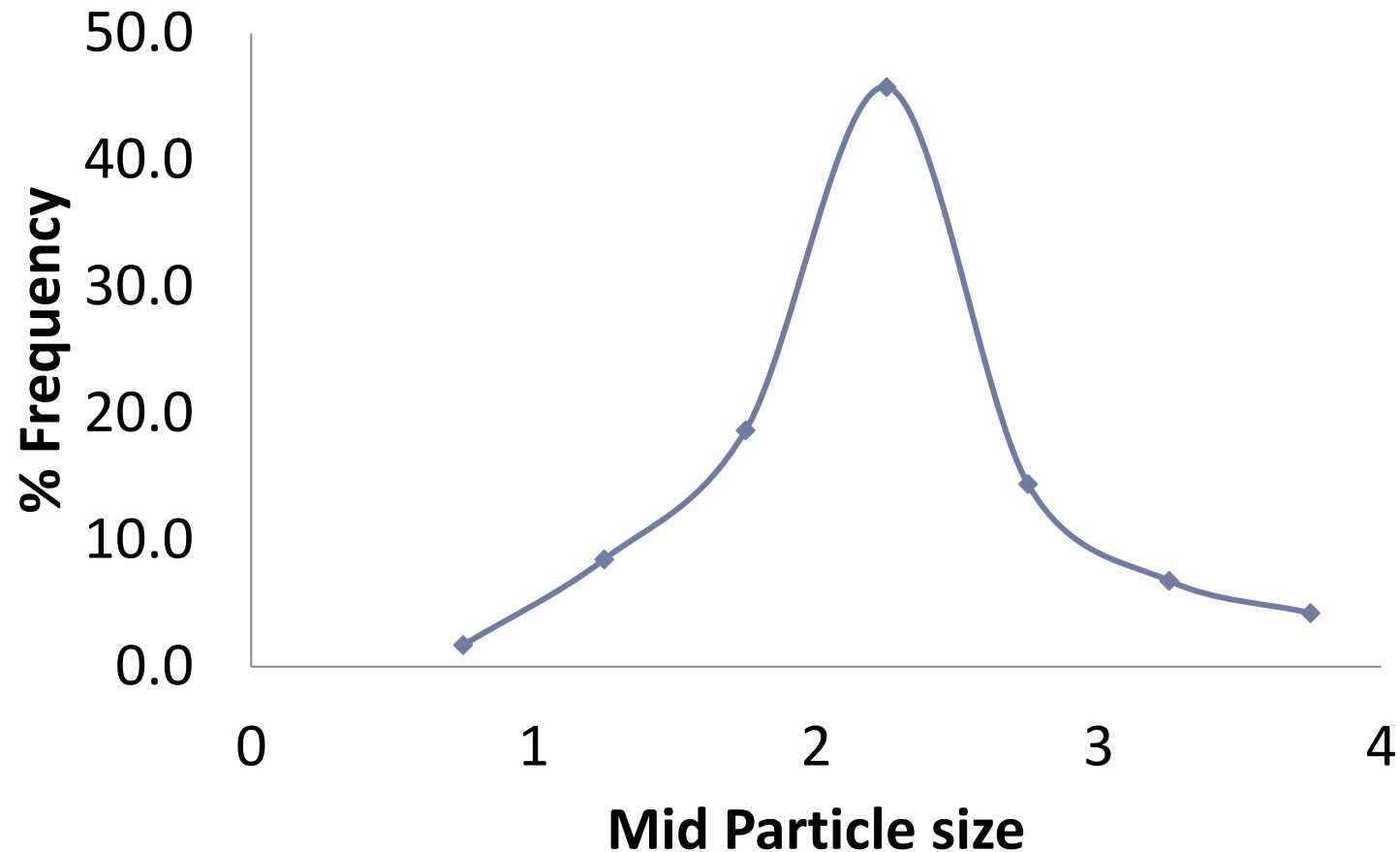
Fundamental Properties ... *Particle size distribution*

Size range (μm)	Mean size range (μm), d	Number of particles in each size range, n	%Frequency	Cumulative % Frequency (undersize)
0.5-1.0	0.75	2	1.7	1.7
1-1.5	1.25	10	8.5	10.2
1.5-2.0	1.75	22	18.6	28.8
2.0-2.5	2.25	54	45.8	74.6
2.5-3.0	2.75	17	14.4	89.0
3.0-3.5	3.25	8	6.8	95.8
3.5-4.0	3.75	5	4.2	100.0
		Sum=118		

Fundamental Properties ... *Particle size distribution*



Fundamental Properties ... *Particle size distribution*



Fundamental Properties ... *Particle size distribution*

► **Normal distribution**

- the distribution is symmetric around the mean, which is also the mode
- The standard deviation, σ , *is an indication of the distribution* about the mean

► **Not commonly found in pharmaceutical powders**

Fundamental Properties ... *Particle size distribution*

- ▶ Pharmaceutical systems tend to have a non-symmetric, or skewed distribution

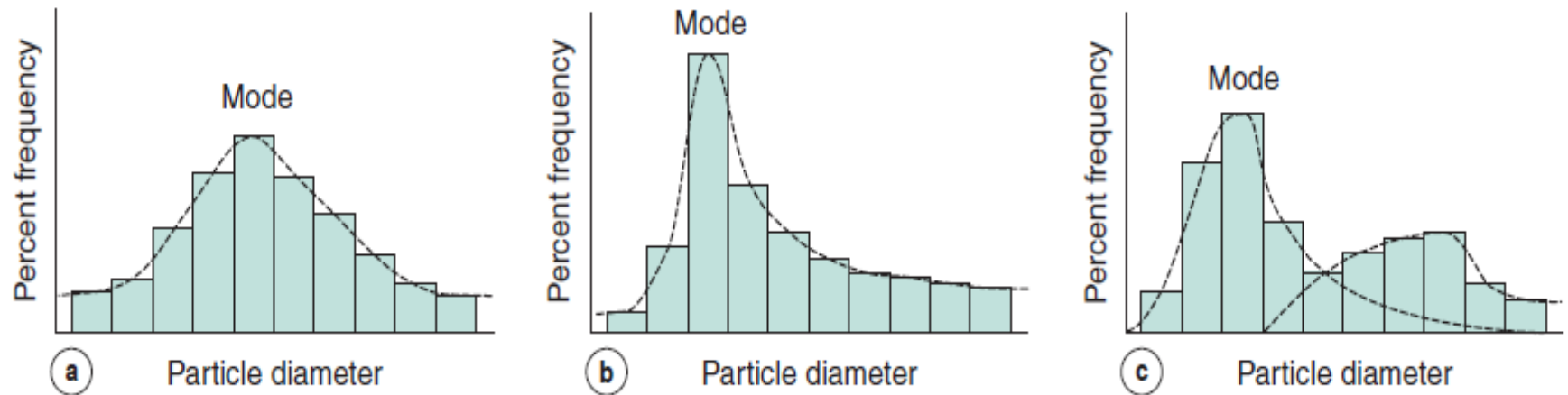
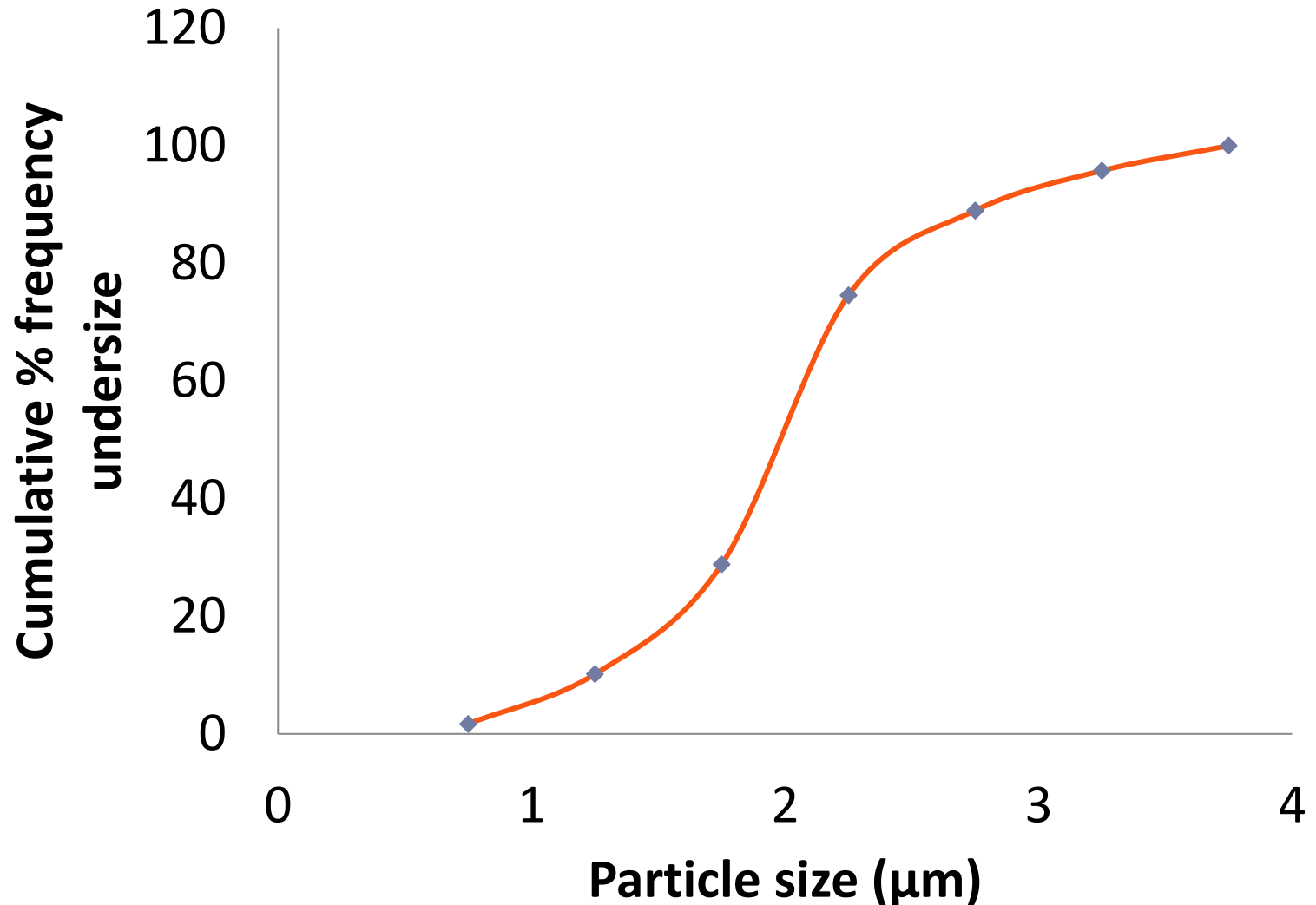
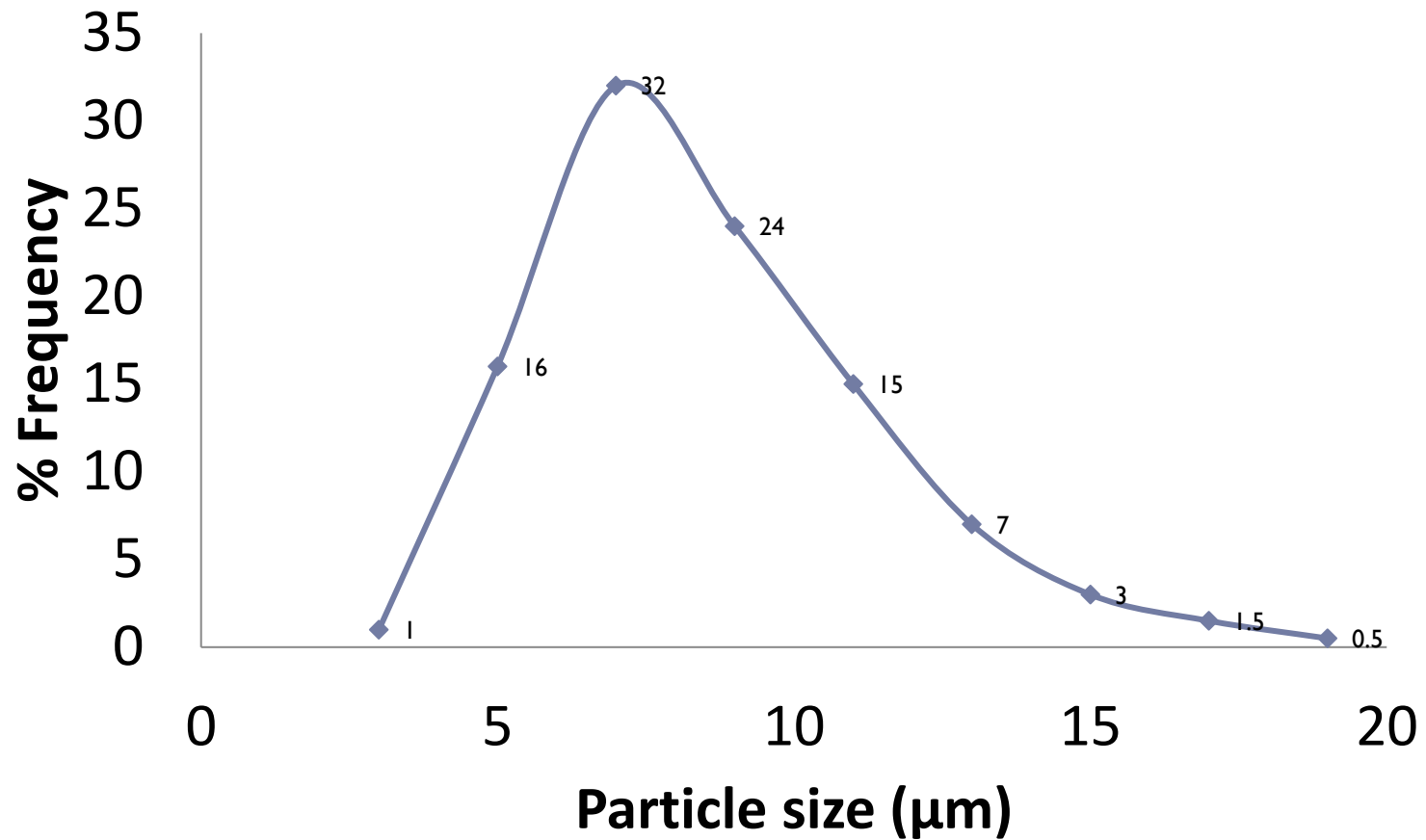


Fig: Size-frequency distribution curves and histograms corresponding to (a) a normal distribution, (b) a positively skewed distribution and (c) a bimodal distribution

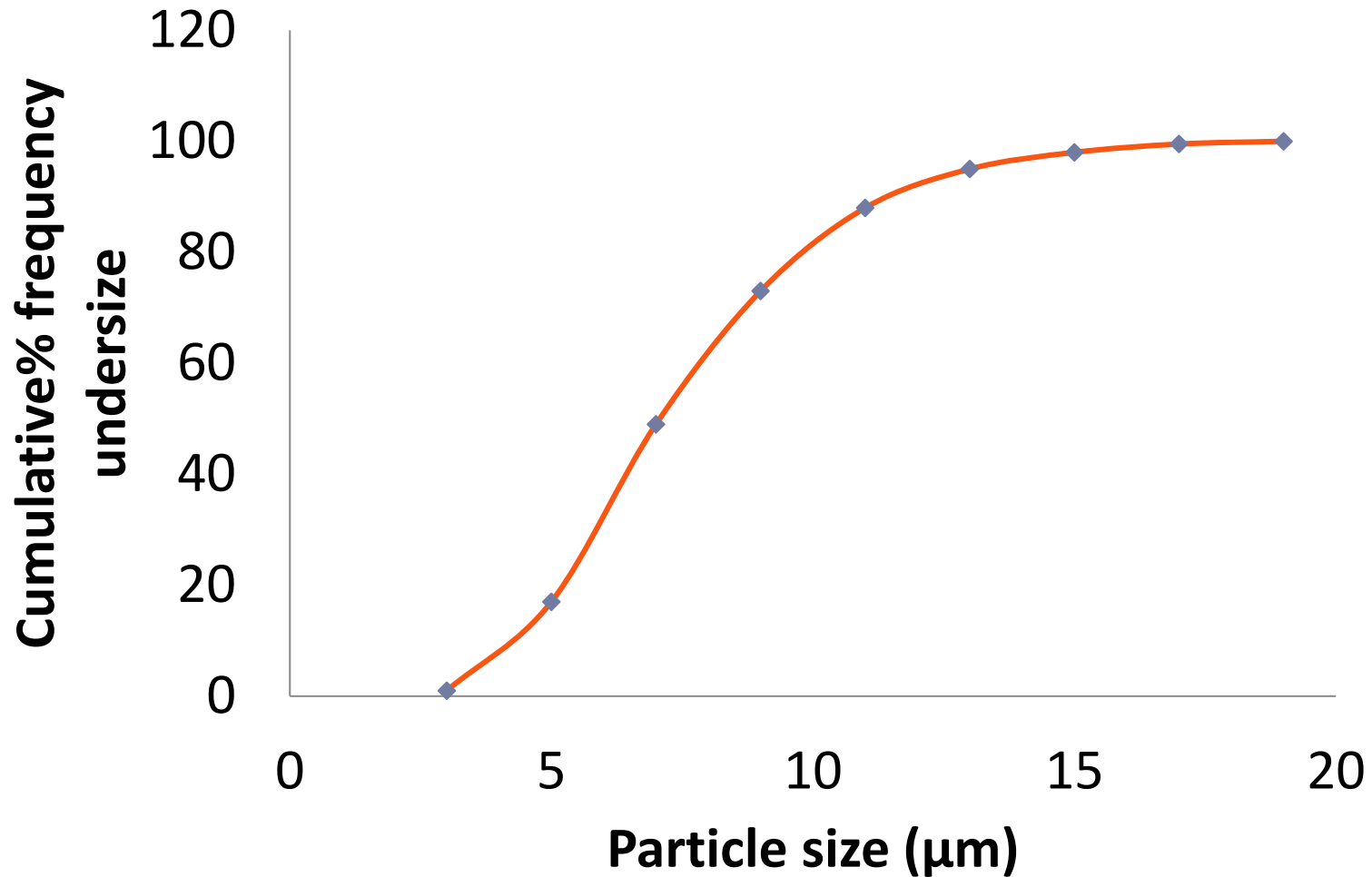
Fundamental Properties ... *Particle size distribution*



Fundamental Properties ... *Particle size distribution*



Fundamental Properties ... *Particle size distribution*



Methods of particle size determination

- ▶ Particle-size analysis methods can be selected and classified based on several criteria:
 - Measuring range required
 - Precision required
 - Quantity of sample to be analyzed
 - Time of analysis
 - Media in which sample can be analyzed (dry or liquid)

Methods of particle size determination...

► Hence,

- Microscopy
- Sieving
- Sedimentation method
- Coulter counter

Microscopy

- The effective size range for analyzing

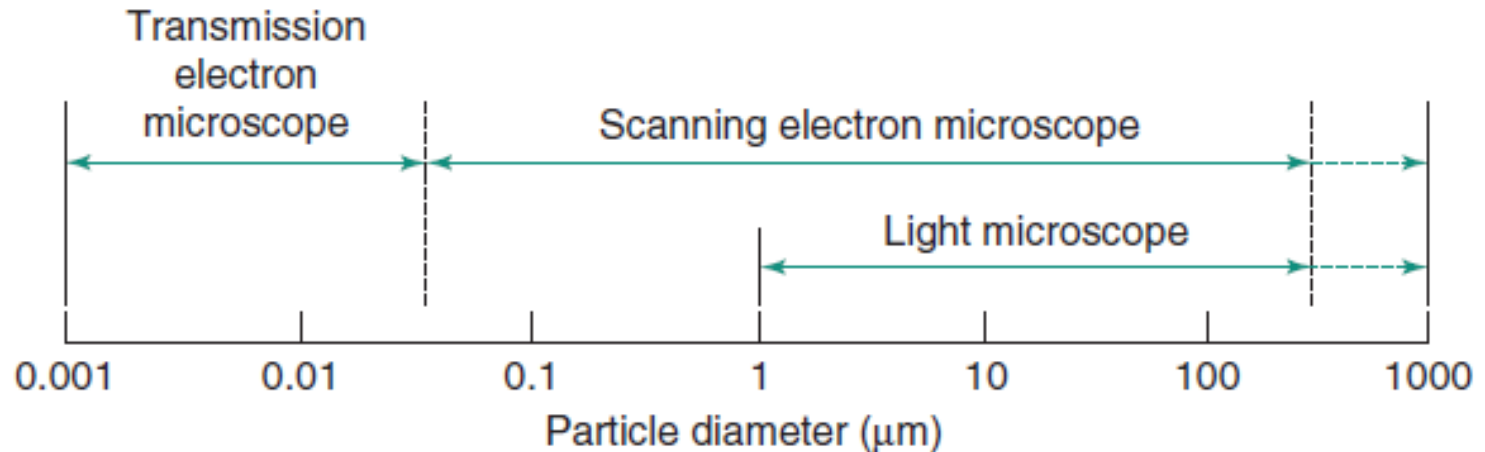


Fig: Size range of analysis using microscopy

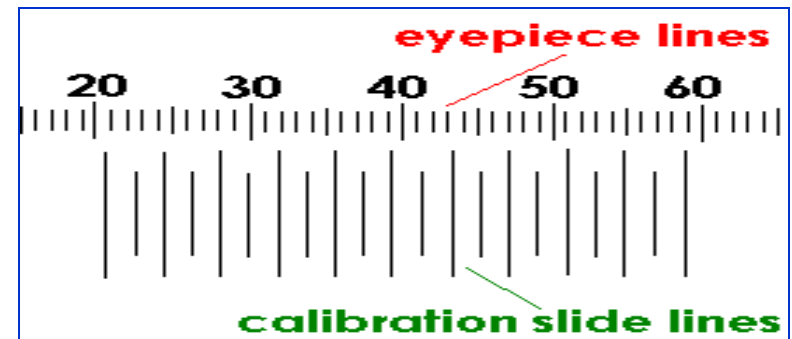
Optical Microscopy

► Procedure

- Dilute suspension of the particles, whose sizes are to be determined, is prepared in a liquid in which it is insoluble
- A drop of suspension is placed on the slide
- The eyepiece of the microscope is fitted with micrometer by which the size of the particles can be determined
- The particles observed are counted
- For ease, the field can be projected or photographed

Optical Microscopy...

- At least **300- 500 particles must be counted** in order to obtain a good size distribution analysis of data

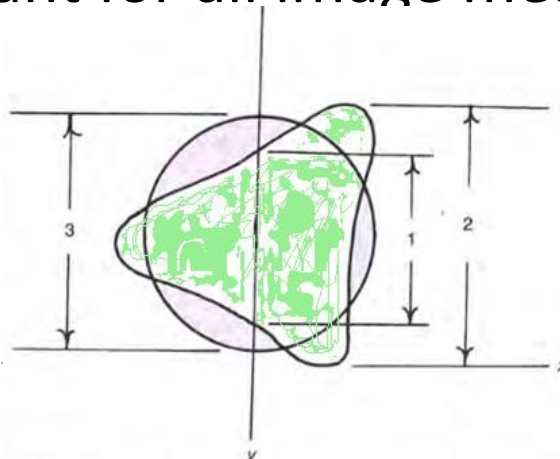


Optical Microscopy...

Diameters measured

Martin's diameter (M) -1

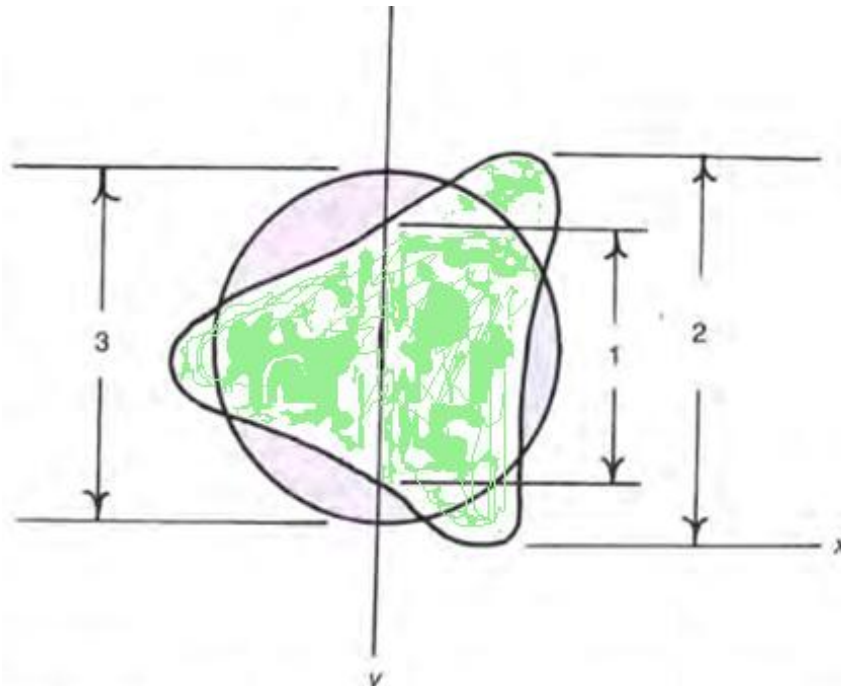
- ▶ The length of the line which bisects the particle image
- ▶ The lines may be drawn in any direction which must be maintained constant for all image measurements



Optical Microscopy...

Feret's diameter (F) -2

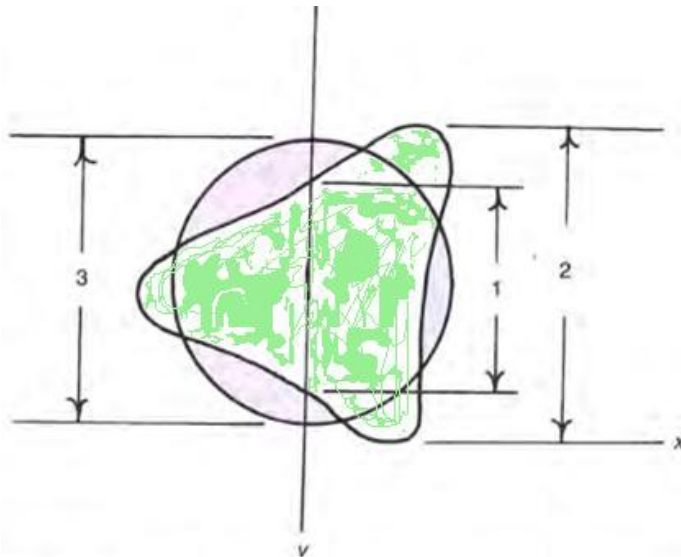
- The distance between two tangents on opposite sides of the particle, parallel to some fixed direction



Optical Microscopy...

Projected area diameter (d_a or d_p)-3

- The diameter of a circle having the same area as the particle viewed normally to the plane surface on which the particle is at rest in a stable position



Optical Microscopy...

Advantages

- Providing a direct visual representation of the particles
- Requires an extremely small amount of sample
- The equipment is relatively inexpensive to acquire and maintain
- Aggregates as well as particles of more than one component can be detected
- It can provide details about shape, crystal habit, and homogeneity within the sample in addition to size

Optical Microscopy...

Disadvantage

- The measured diameter of the particles represents two dimensions only
- Slow and tedious process

Sieve Analysis

- For measuring of the size of coarse powder

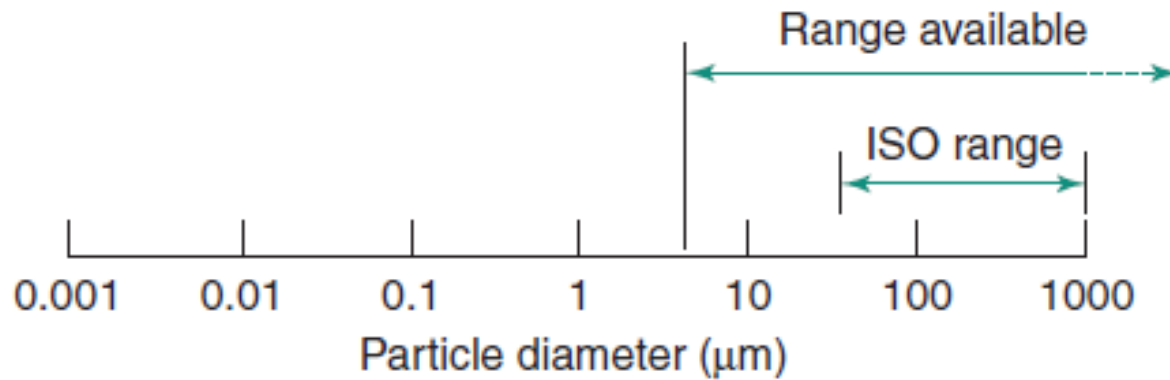
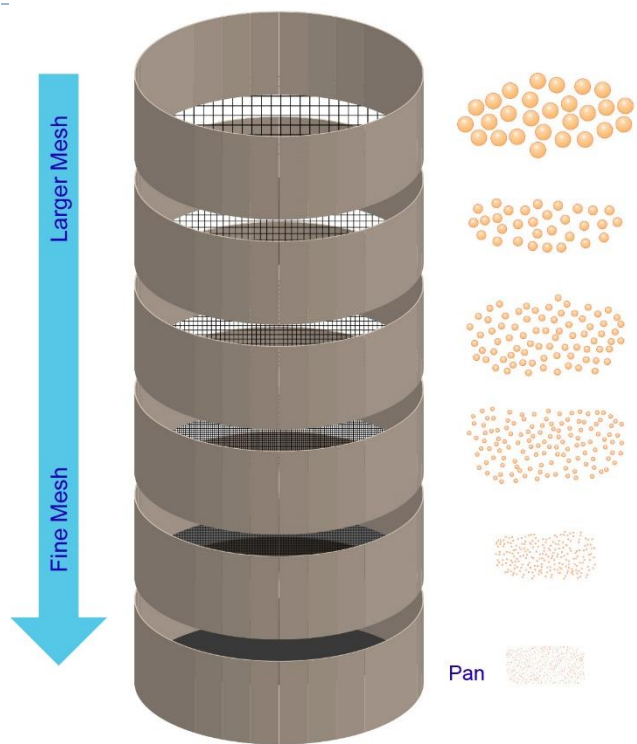


Fig: Size range of analysis using sieves. ISO, International Organization for Standardization



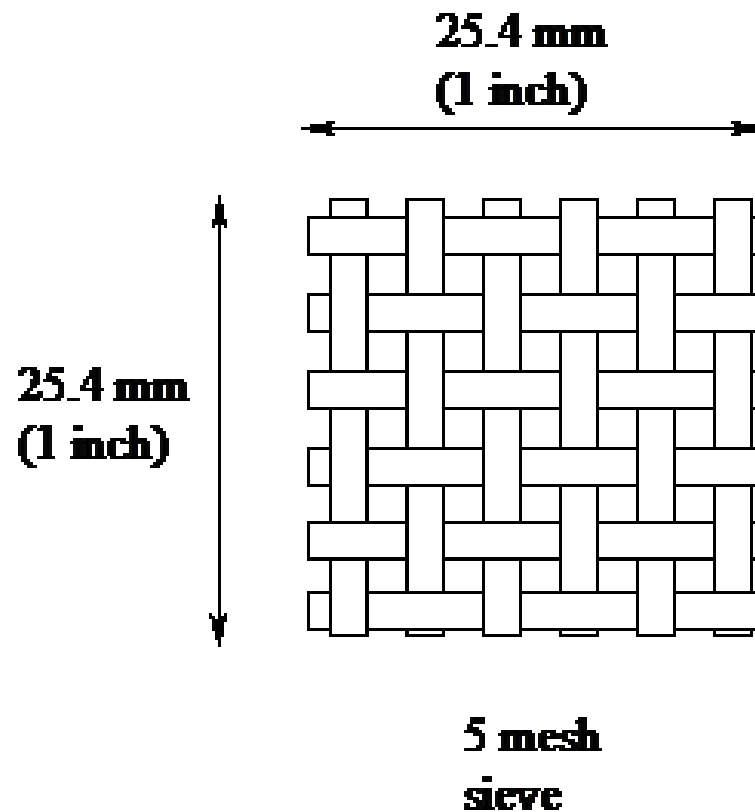
Sieve Analysis...

► It is used for

- ✓ Classification of powders
- ✓ Determining the size distribution of coarse powders
- Sieves are expressed in number
- The number of the sieve indicates
 - ✓ the number of meshes included in a length of 25.4 mm (1 inch) in each direction parallel to the wires

Sieve Analysis...

- ▶ A No. 5 sieve has 5 meshes per inch in each direction



Sieve Analysis...

Table Openings of Standard Sieves,
U.S. Series

Assigned number	Sieve opening
2	9.5 mm
3.5	5.6 mm
4	4.75 mm
8	2.36 mm
10	2.00 mm
14	1.40 mm
16	1.18 mm
18	1.00 mm
20	850 μm
25	710 μm
30	600 μm
35	500 μm
40	425 μm
45	355 μm
50	300 μm
60	250 μm
70	212 μm
80	180 μm
100	150 μm
120	125 μm
200	75 μm
230	63 μm
270	53 μm
325	45 μm
400	38 μm



Sieve Analysis...

► Pharmaceutical powders classification by fineness

Classification of Powder	d_{50} Sieve Opening (μm)
Very Coarse	> 1000
Coarse	355–1000
Moderately Fine	180–355
Fine	125–180
Very Fine	90–125

Sieve Analysis...

- ▶ A nest of sieves with the coarsest on top is placed on the shaker, and the powder sample of **known weight** is placed on the top of the sieve & shaken for a **definite period of time**
- ▶ The material that passes through one sieve and is retained on the next finer sieve is collected and weighed



Advantages & Disadvantage of Sieving method

► Advantage

- Simple
- Low cost
- Little skill required for operation

Advantages & Disadvantage of Sieving method

► Disadvantage

- Large sample size required
- Blinding of the screens due to mechanical clogging
- Aggregation- due to electrostatic charge or moisture
- Attrition- size reduction



Sedimentation method

- ▶ The sedimentation method employs the settling of particles in a liquid of a relatively low density,
 - under the influence of a gravitational or centrifugal field

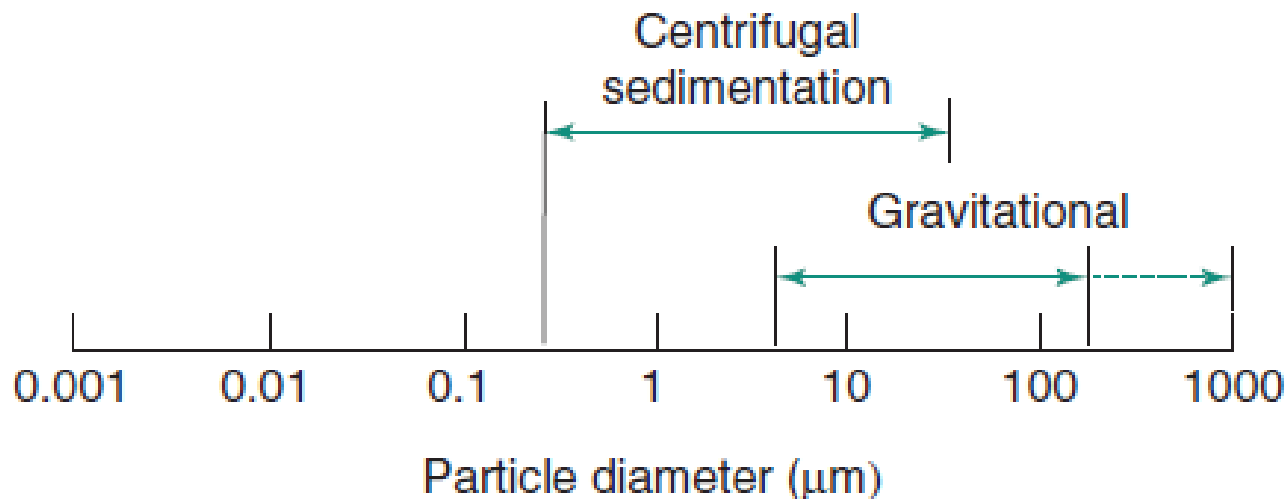


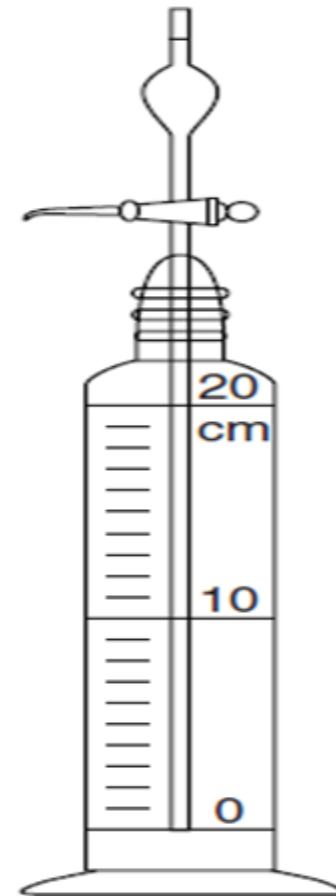
Fig: Size range of analysis using sedimentation methods

Sedimentation method...

- ▶ Several methods based on sedimentation are used
 - The *Andreasen pipette*, which is based on sampling near the bottom of a **glass sedimentation chamber**, is the best known one
 - Because it combines **ease** of analysis, **accuracy** and **economy** of equipment

Sedimentation method...

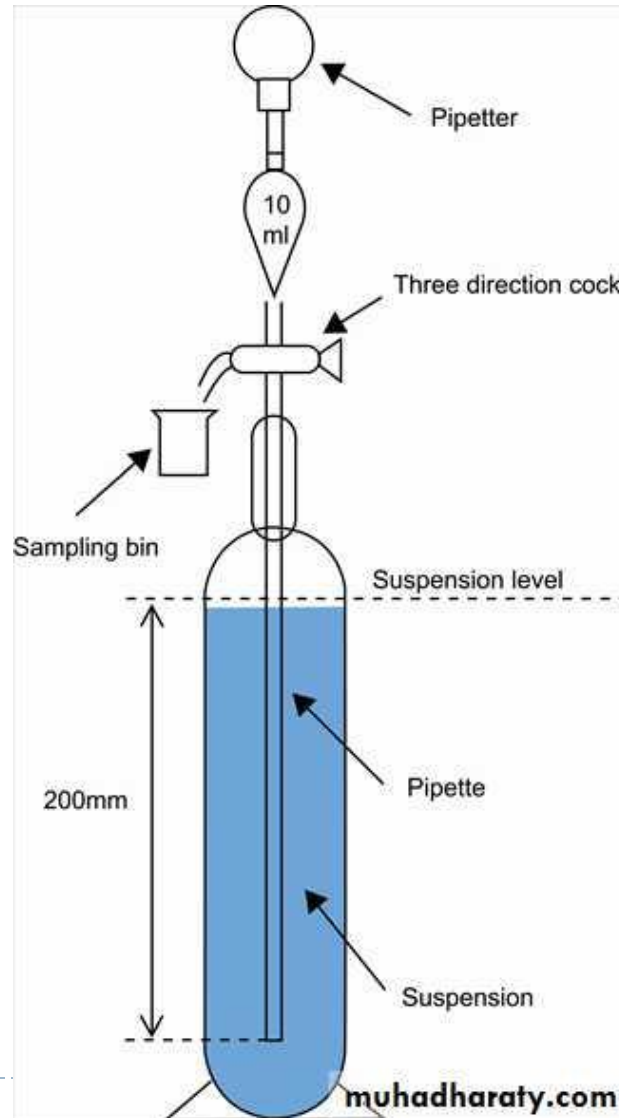
- ▶ **Andreasen pipette** is used for particle size distribution determination
- 550 ml stoppered cylindrical vessel with 5.5 cm internal diameter
- The stopper has an integral 10 ml bulb pipette
- ✓ Its lower tip should be 20 cm below the surface of the suspension



Sedimentation method...

Size distribution is determined as follows:

- Allow a homogeneous suspension(1-2%) placed in the vessel up to 550 ml mark to settle in a cylinder
- Left undisturbed in constant temperature bath
- Take 10 ml sample from the settling suspension at a fixed horizontal level at intervals of time



Sedimentation method...

- Evaporate and **weigh** the samples
- This concentration expressed as a percentage of the initial concentration gives the percentage (w/w) of particles whose falling velocities are equal to or less than **x/t**
- Substitution in the equation gives the corresponding Stokes' diameter

Sedimentation method...

- The particle size can be obtained by gravity sedimentation as expressed in Stokes' law

- **Stokes' Law**

$$v(x/t) = \frac{(\rho_s - \rho_o) g d_{st}^2}{18\eta_o}$$

$$d_{st} = \sqrt{\frac{18\eta_o}{(\rho_s - \rho_o) g} \frac{x}{t}}$$

Sedimentation method...

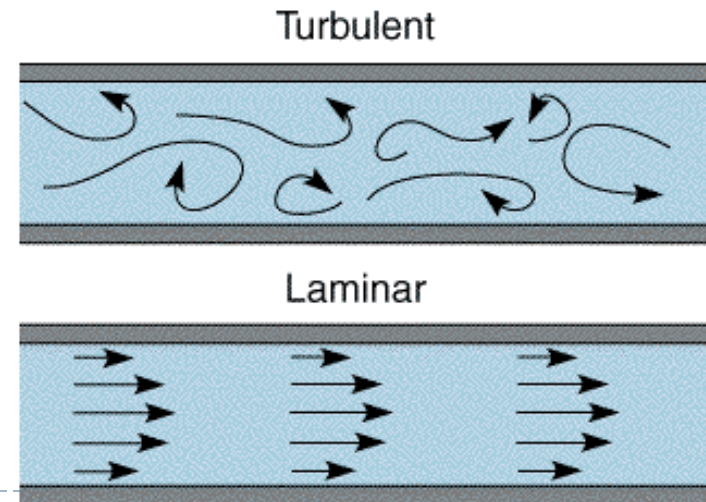
Example

- A sample of powdered *zinc oxide*, density 5.60 g/cm^3 is allowed to settle under the acceleration of gravity, 981 cm/sec^2 at 25°C . The rate of settling v is $7.30 \times 10^{-3} \text{ cm/sec}$; the density of the medium is 1.01 g/cm^3 , and its viscosity is 1 centipoise = 0.01 poise or 0.01 g/cm sec . Calculate the Stokes' diameter of the zinc oxide Powder.

$$d_{st} = \sqrt{\frac{(18 \times 0.01 \text{ g/cm sec}) \times (7.3 \times 10^{-3} \text{ cm/sec})}{(5.60 - 1.01 \text{ g/cm}^3) \times (981 \text{ cm/sec}^2)}}$$
$$= \underline{\underline{5.40 \times 10^{-4} \text{ cm}}} \quad \text{or} \quad \underline{\underline{5.40 \mu\text{m}}}$$

Sedimentation method...

- ▶ For Stokes' law to apply, a further requirement is that the **flow of dispersion medium** around the particle as it sediments is ***laminar or stream line***.



Sedimentation method...

Advantages

- The apparatus is inexpensive and the technique is simple
- The results obtained are precise provided the technique is adequately standardized

Sedimentation method...

Disadvantages

- Method is laborious since separate analysis are required for each experimental point on the distribution curve
- Very small particles cannot be determined accurately since their settling is unduly prolonged

Coulter counter method

- ▶ Equivalent volume diameter is measured

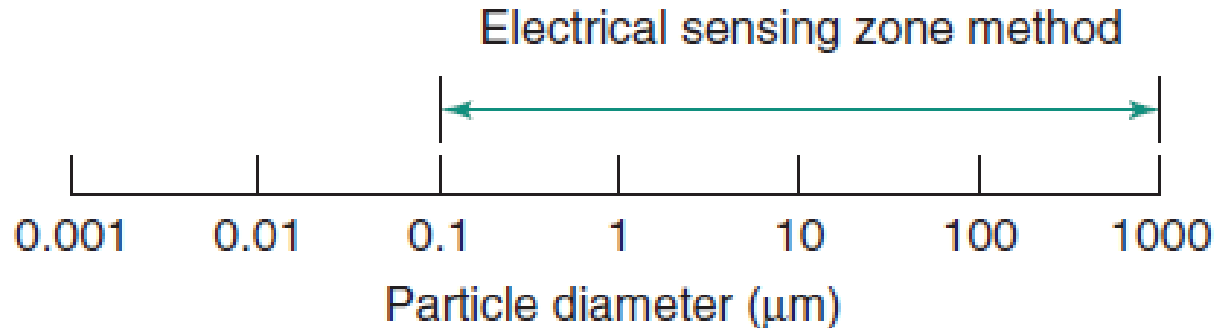


Fig: Size range of analysis using the electrical sensing zone method

Coulter counter method...

► Principle:

- when a particle suspended in a conducting liquid passes through a small orifice (opening), on either side of which are electrodes, a change in electric resistance occurs
- Powder samples are dispersed in the electrolyte to form a very dilute suspension
- A known volume of suspension is pumped through the orifice so that only one particle passes at a time through the orifice

Coulter counter method...

- A constant voltage is applied across the electrodes so as to produce a current
- As the particle travels through the orifice, it displaces its own volume of electrolyte and this results in an increased resistance b/n the two electrodes
- The change in the electrical resistance b/n the electrodes that is proportional to the volume of the particle is converted in to a voltage pulse that is amplified and processed electronically

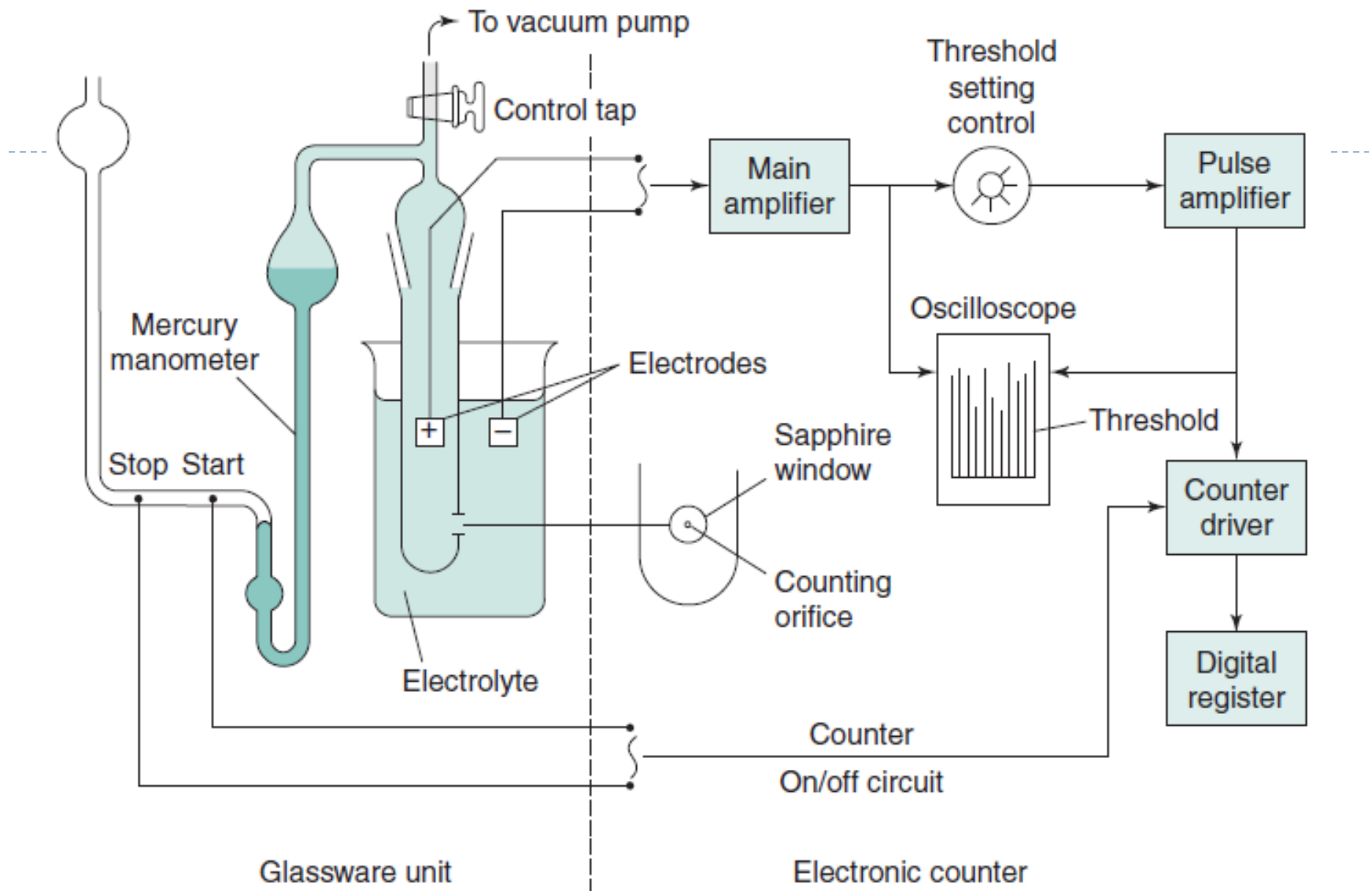


Fig: Electrical sensing zone apparatus

Coulter counter method...

► Advantages

- Evaluation is fast
- Since a large number of particles are counted, the results are more reliable
- Since the aperture is automatic, operator variability is avoided

Coulter counter method...

► Disadvantages

- The sample must be suspended in an electrolyte
 - ✓ In many cases the fluid in which the particles are suspended is non-conducting
- Aggregation of particles can give false results

PARTICLE SHAPE AND SURFACE AREA



Introduction

- ▶ Knowledge of the shape and the surface area of a particle is desirable
- ▶ Particle shape influences
 - Surface area
 - Flow property
 - Packing
 - Compaction properties of particles

Introduction...

- ▶ The surface area per unit weight or volume is an important characteristic of a powder
 - Surface adsorption and dissolution rate studies

Surface area determination

- ▶ Surface area determination
 - Adsorption method
 - Fluid permeability methods

Adsorption method

- ▶ Working principle

- Measures the adsorption of gas molecule

- ✓ Degree of adsorption is proportional to SA

- ▶ Most substances adsorb **mono molecular layer** of gas at a given temperature

- ▶ The volume of n gas adsorbed at monomolecular per gram of powder is given by **Brunauer Emmett Teller (BET)**

equations

Adsorption method...

► BET equation

$$\frac{p}{V(p_0 - p)} = \frac{1}{V_m b} + \frac{(b - 1)p}{V_m b p_0}$$

► Where

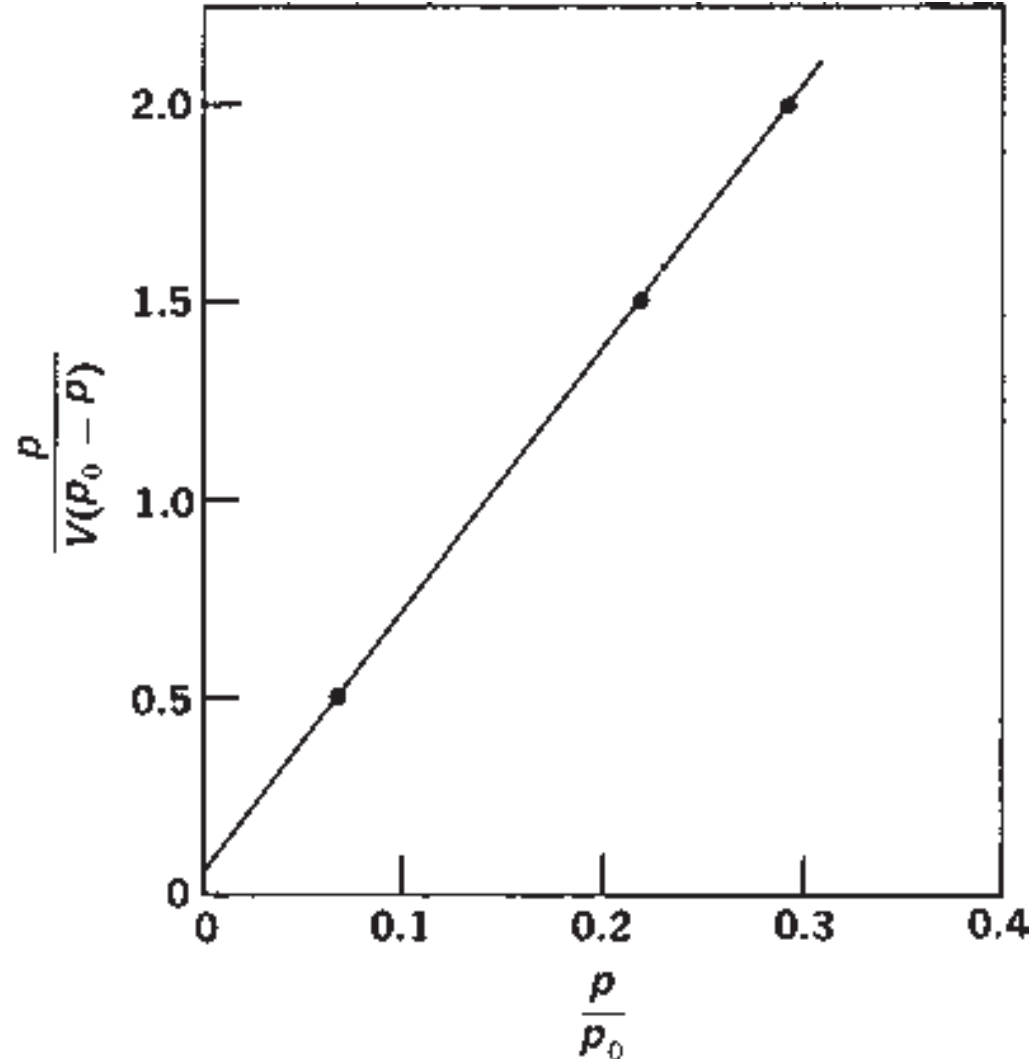
- V is the volume of gas in cm^3 adsorbed per gram of powder at pressure p
- P_0 is the saturation vapor pressure of liquefied nitrogen at the temperature of the experiment
- b is constant
- V_m is the volume of gas required to cover unit mass of adsorbent with monolayer

Adsorption method...

► Plot of $P/V (P_o - P)$ Vs P/P_o

— Slope = $b - 1/V_m b$

— Intercept = $1/V_m b$



Adsorption method...

► The total SA

$$S_w = \frac{A_m N}{M/\rho} \times V_m \text{ cm}^3/\text{g}$$

- N- Avogadro's number (6.02×10^{23} molecules/mole)
- A_m – cross sectional area of adsorbate ($16.2 \times 10^{-16} \text{ cm}^2$)
- M/p- is the molar volume of the gas ($22,414 \text{ cm}^3/\text{mole}$ at STP)

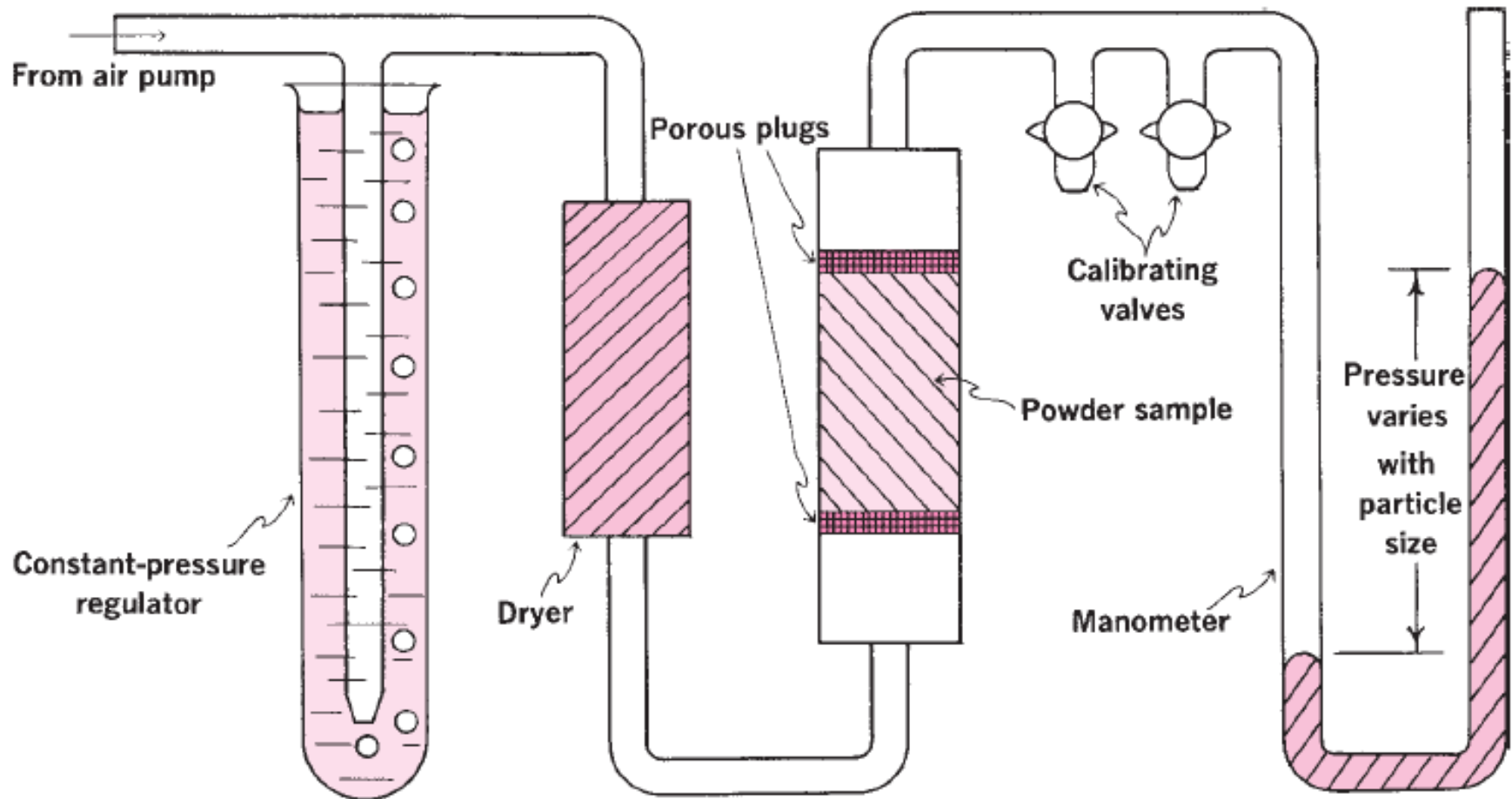
-
- Calculate the specific surface of the powder, in m^2/g , for the given data:

$p/V(p_0 - p)$	0.05	0.150	0.20
p/p_0	0.07	0.220	0.290

Air permeability method...

- The principle: based on the resistance to the flow of a fluid, such as air, through a plug of compacted powder
- Rate of flow that is permeability of fluid depends on particle size
- The greater the surface area per gram of powder, the greater the resistance to flow
- Measurement of rate of fluid enables mean particle size to be calculated

Air permeability method...



Air permeability method...

$$V = \frac{A}{\eta S_w^2} \cdot \frac{\Delta P t}{K l} \cdot \frac{\varepsilon^3}{(1 - \varepsilon)^2}$$

A- cross-sectional area of the plug

K- is a constant that takes account of the irregular capillaries

ε - is the porosity

l- length of the capillary

ΔP - pressure difference

t-seconds

S_w - specific surface area of the particles

V- volume of air flowing through a capillary

Derived Properties

Derived properties of powders

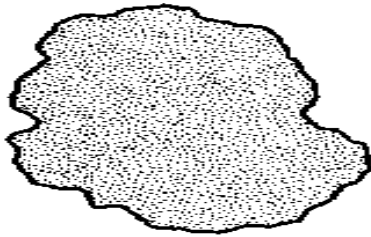
- Porosity
- Density
- Bulkiness or specific bulk volume
- Flow properties

Porosity of powder

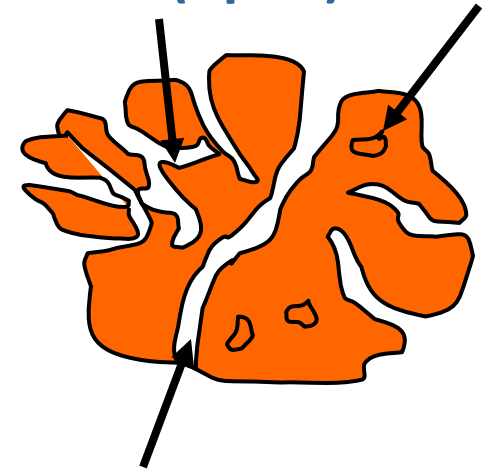
- ▶ The quality or state of being porous or the proportion of a powder bed or compact that is occupied by pores
 - **It is a measure of the packing efficiency of a powder**
- ▶ Powders can be
 - i. **Porous** (most pharmaceutical solids are porous, i.e., they have internal pores or capillary)
 - ✓ **Bulk volume > true volume**

Porosity of powder...

II. Non-porous



Inter-connected (open) Closed



Passing (open)

Porosity of powder...

► Bulk volume

- ✓ When a powder is placed in a graduated cylinder
 - ✓ the total volume occupied is known as the bulk volume V_b

► For non-porous powder

- ✓ the bulk volume of a powder consists of true volume of the solid particles plus volume of the spaces between particles

Bulk volume (V_b) = True volume (V_p) + Volume of spaces b/n particles

Porosity...

➤ The volume of the spaces, the **void volume**,

$$V = V_b - V_p$$

➤ **The porosity (ϵ)** of powder is determined as the ratio of void volume to bulk volume

$$\epsilon_{Tot} = \frac{V_b - V_p}{V_b} = 1 - \frac{V_p}{V_b} = \epsilon_{Inter} \text{ (for non porous)}$$

Frequently expressed in percent, $\epsilon \times 100$

Porosity...

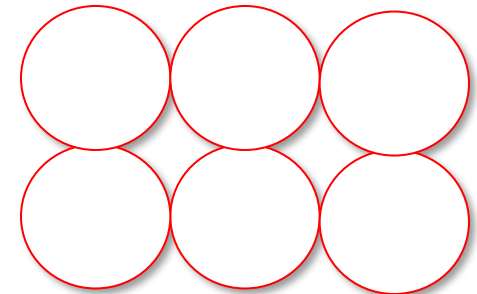
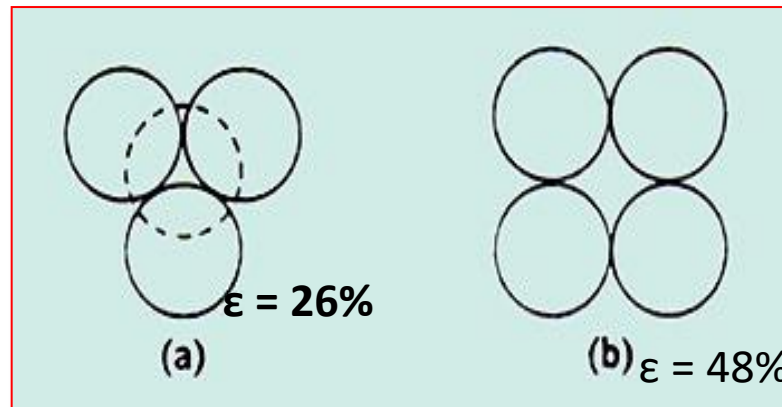
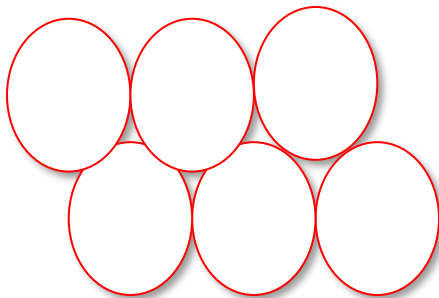
Packing arrangement in powder beds

- ▶ Two types of packing are possible for spherical particles

Rhombohedral packing

Cubic packing

- Closest packing ($\epsilon=26\%$) Most open/ Loosest packing ($\epsilon=48\%$)



Porosity...

- ▶ When the particles of varying sizes are present, porosity lower than the theoretical minimum of 26% is also possible. **Why ?**
- ▶ If the powder contains floccules or aggregates, the porosity may go beyond the theoretical maximum of 48%. **Why ?**
- ▶ Highly compressed crystalline materials, $\varepsilon < 1\%$

Porosity...

► for porous powder:

$$\varepsilon_{\text{intra}} = \frac{V_g - V_p}{V_g} = 1 - \frac{V_p}{V_g} = 1 - \frac{\rho_g}{\rho_p}$$

$$\begin{aligned}\varepsilon_{\text{interspace}} &= \frac{V_b - V_g}{V_b} = 1 - \frac{V_g}{V_b} \\ &= 1 - \frac{\text{Weight/Granule density}}{\text{Weight/Bulk density}}\end{aligned}$$

$$\begin{aligned}\varepsilon_{\text{interspace}} &= 1 - \frac{\text{Bulk density}}{\text{Granule density}} \\ &= 1 - \frac{\rho_b}{\rho_g}\end{aligned}$$

Note: for nonporous powder $V_g = V_p$

Density of powder...

Example:

- ▶ Estimate the Intraparticle porosity of sulfadiazine granules having a granule density of 1.12 g/cm³ and true density of 1.5g/cm³.

$$\begin{aligned}\epsilon_{\text{intra}} &= \frac{V_g - V_p}{V_g} = 1 - \frac{V_p}{V_g} \\ &= 1 - \frac{\rho_g}{\rho_p}\end{aligned}$$

- ▶ Ans=25.3%

Examples on porosity

Example

- ▶ A sample of calcium oxide powder with a true density of 3.203 and weighing 131.3 g was found to have a bulk volume of 82 cm³ when placed in a 100-ml graduated cylinder. Calculate the porosity ? **Ans.=50%**
- ▶ Calculate the percent porosity of TiO₂ having a true density of 4.26g/cm³ and 100 g sample of which was found to occupy a bulk volume of 80 mL. **Ans.=70%**

Density of powder

- ▶ Density is defined as weight per unit volume (W/V)
 - Difficulty arises when one attempts to determine the volume of particles containing microscopic cracks, internal pores, and capillary spaces
- ▶ Based on the method of determination, three types of densities can be defined
 - the **true density, granule density and bulk density**

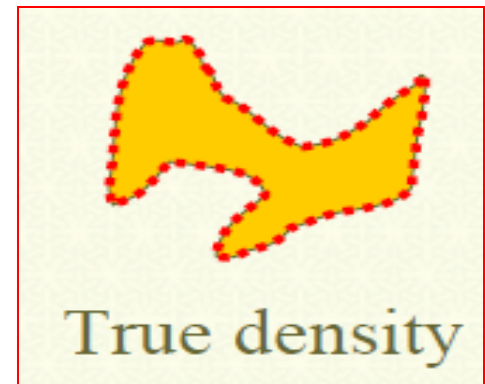
Density of powder...

A- True density

- ▶ The true density, or absolute density, of a sample excludes the **volume of the pores** and **voids within the sample**

Methods

- ✓ Liquid displacement method
- ✓ Gas displacement method (He, H₂)
 - better penetration ability



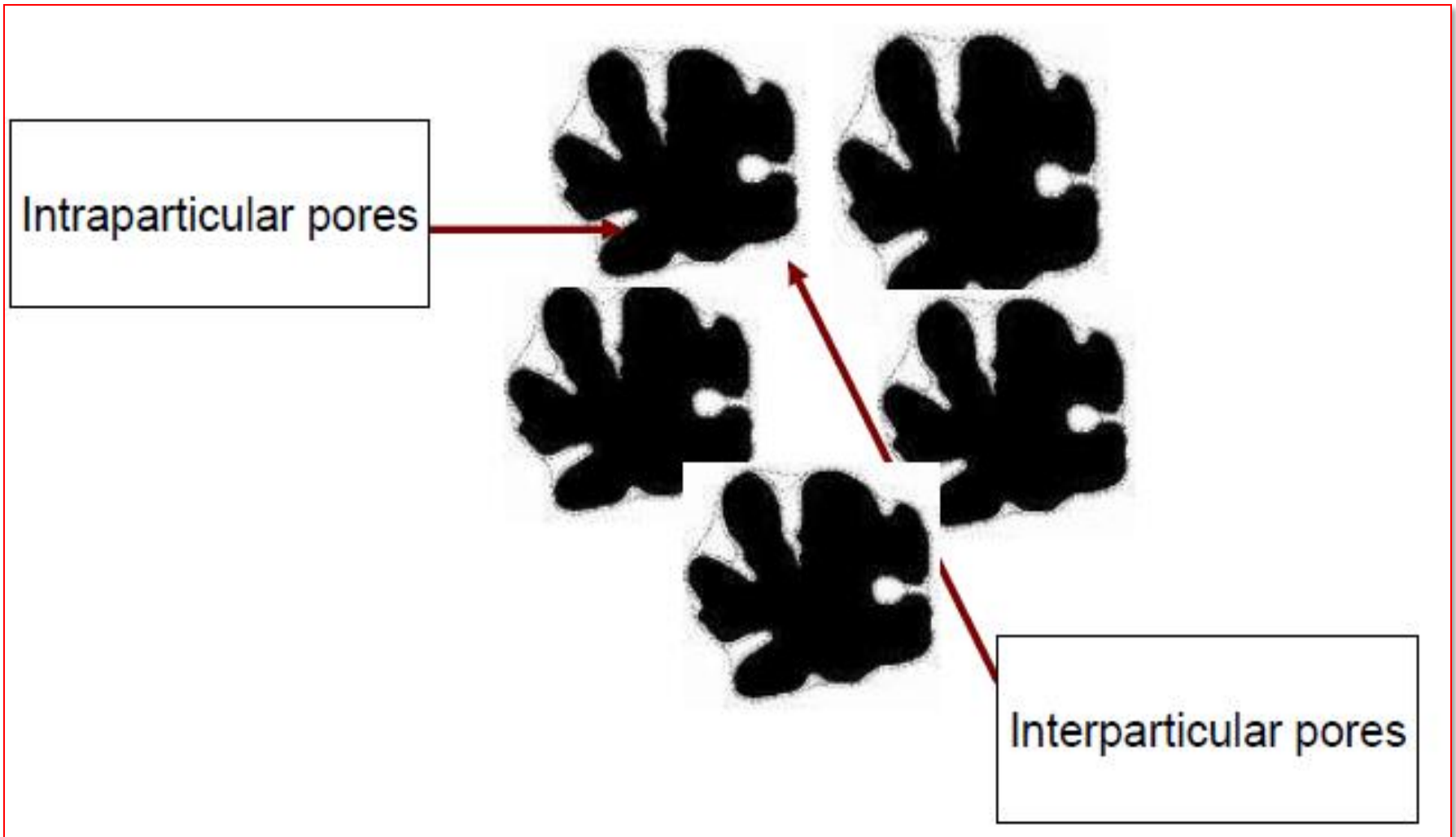
Density of powder...

B. Granule density (ρ_g)

- Mass of the granular powder and the volume occupied by the granular material together with its intra particle space
 - **Method-using Liquid displacement method (Mercury)**



Density of powder...



Density of powder...

C- Bulk density (ρ_b)

- It is the ratio of the mass of the powder and its bulk volume
- Includes the volume of all of the pores within the sample



Density of powder...

- A sample of powder is carefully introduced into a 100 mL graduated cylinder
- The cylinder is dropped onto a hard wood surface three times from a height of 1 inch
- The bulk density is then obtained by dividing the weight of the sample in grams by the final volume in cm^3 of the sample contained in the cylinder

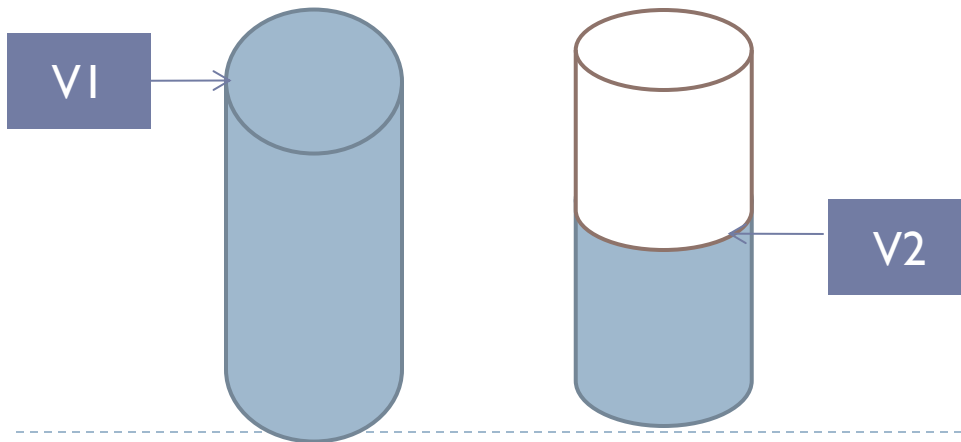
Density of powder...

C2. Tapped density (ρ_T)

► It is the ratio of mass of powder to tapped volume

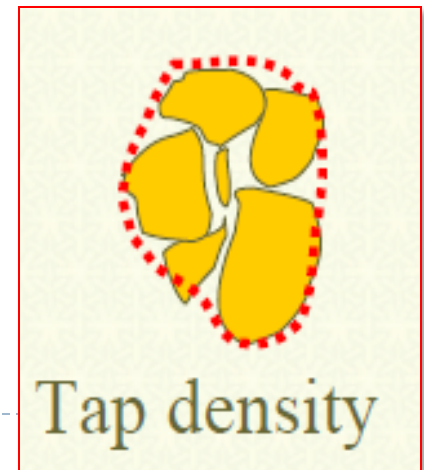
—Tapped volume

✓ the volume of the powder obtained after tapping the powder 250 times using **Tap Densitometer**



$$\rho_b = \frac{M}{V_1}$$

$$\rho_T = \frac{M}{V_2}$$



Density of powder...



Tap densitometer

Bulkiness or specific bulk volume

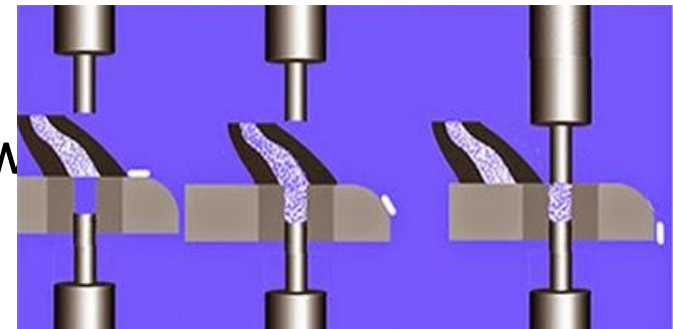
- ▶ It is the reciprocal of bulk density
- ▶ **Bulkiness usually increases with a decrease in particle size**
 - However, in a mixture of particles with different sizes, the bulkiness may get reduced

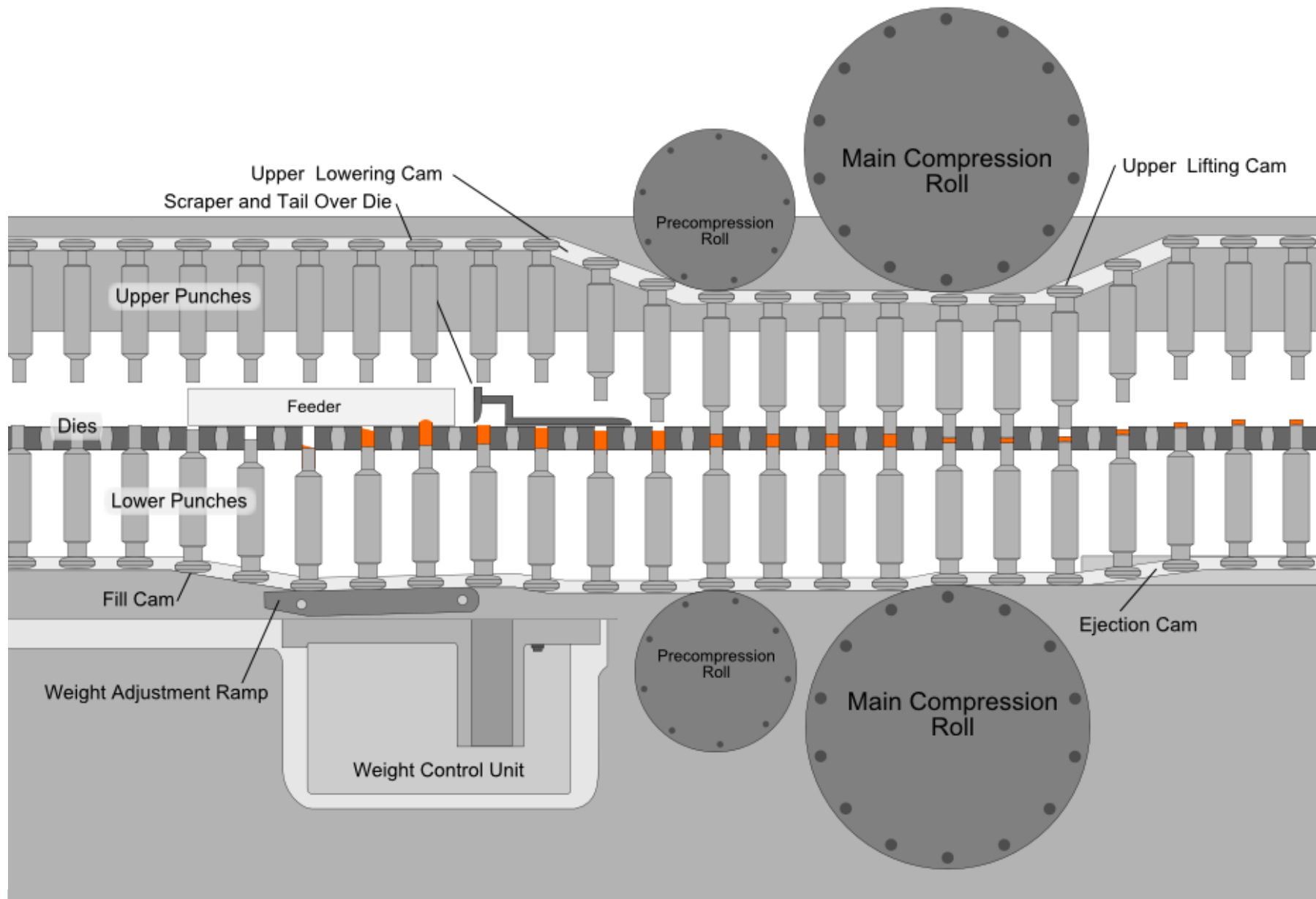
Application of Bulkiness

- It is a useful property to be considered while **choosing a suitable container for packaging or during filling of drug powders in to capsules**

Flow properties of powders

- Important parameter to be considered in the production of pharmaceutical dosage forms
- **Example:**
 - ✓ Uniform filling of dies during tableting, and proper filling of capsules during capsule filling
 - ❖ directly depends on the flow properties of the powder







Flow properties of powders...

- ▶ Based on their flowability, the pharmaceutical powders are classified as:
 - Free-flowing : easily flowing
 - Cohesive (“sticky”) or non-free-flowing: do not flow as readily as possible

Flow properties of powders...

The poor flow in powders is attributed to:

- 1) Cohesiveness or stickiness between particles** due to presence of **Van der Waals, and electrostatic forces**
 - ❑ Cohesiveness of particles has been found to depend upon a number of factors
 - ❖ Density or porosity of the powders
 - ✓ Dense materials tend to be less cohesive than lighter ones

Flow properties...

- ❖ The presence of adsorbed materials on the powder surface
 - ✓ Moisture increase cohesiveness of particles
- 2) Adhesion between the particles and the container wall due to the above forces
- 3) Friction between particles due to surface roughness
- 4) Physical interlocking of particles specially if these are of irregular shape

Measuring flow property of powders

- ▶ **Flow properties of powders can be assessed by**
 - Compressibility index and Hausner ratio
 - Angle of repose
 - Flow rate

Measuring flow property...

► Compressibility index (Carr's Index) and Hausner ratio

—The compressibility index (CI) and Hausner ratio are calculated from bulk and tapped densities, using Equation

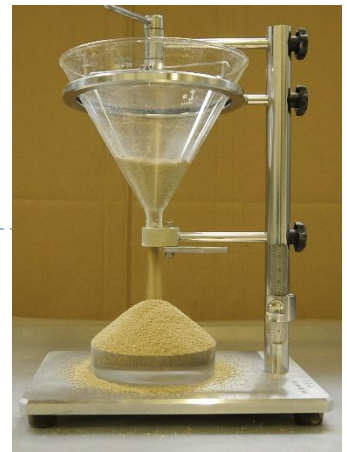
$$\text{Compressibility Index} = \frac{(\rho_t - \rho_b)}{\rho_t} \times 100$$

$$\text{Hausner ratio} = \frac{\rho_t}{\rho_b}$$

Measuring flow property...

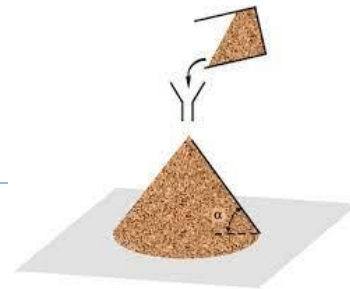
Compressibility Index (%)	Flow Character	Hausner Ratio
≤ 10	Excellent	1.00–1.11
11–15	Good	1.12–1.18
16–20	Fair	1.19–1.25
21–25	Passable	1.26–1.34
26–31	Poor	1.35–1.45
32–37	Very poor	1.46–1.59
>38	Very, very poor	>1.60

Measuring flow property...



► Angle of repose

- Angle of repose is a function of the **surface roughness**
 - ❖ The rougher and more irregular the surface of particles, the more the angle of repose
- As the particles become less and less spherical, the angle of repose increases while the bulk density and flowability decreases



Measuring flow property...

- A sample of powder will be made to flow through a funnel, on to a graph paper from a certain height
- The height (h) as well as the diameter (d) of the pile will be measured
- ❖ Then the angle of repose (θ) was calculated from h and radius ($r = d/2$) using Equation

$$\tan \theta = h / r$$

$$\theta = \tan^{-1}(h / r)$$



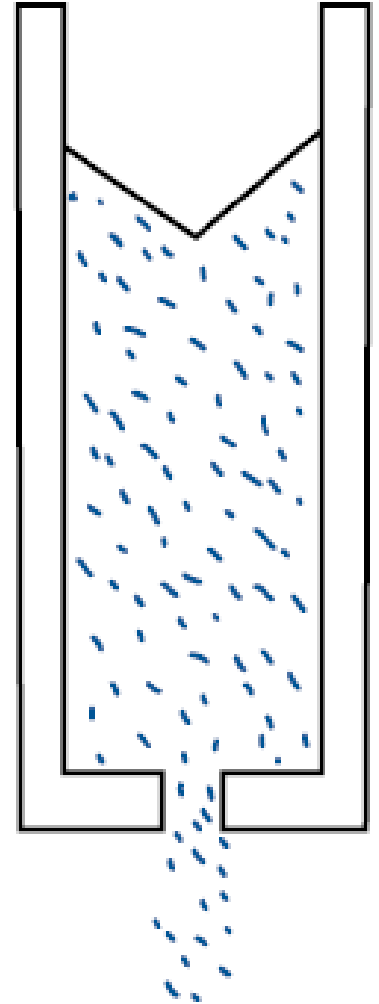
Measuring flow property...

Angle of repose (θ)	Flow properties
$<25^\circ$	Excellent
$25 - 30^\circ$	Good
$30 - 40^\circ$	Satisfactory
$40 - 50^\circ$	Poor
$>50^\circ$	Very poor

Measuring flow property...

► Flow rate

- The flow rate through an orifice is measured as the **mass per time** flowing from any of a number of types of containers
- **Useful for only free flowing materials**



Improvement of the flow properties of powders

1. Alteration of particle's size and distribution

— ↑ particle size → improves flow property

✓ Due to reduction in cohesive forces

— Coarse particles are more preferred than fine ones as they are less cohesive



Improvement...

2. Alteration of particle shape and texture

- Particle's shape: generally, more spherical particles have better flow properties than more irregular particles
- Spherical particles are obtained by **granulation** techniques
- Particle's texture:
 - ✓ Particles with very rough surfaces will be **more cohesive** and have **a greater tendency to interlock** than smooth surfaced particles

Improvement...

3. Removing extra moisture

- Moisture content of particle greatly affects powder's flowability
 - Drying the particles will reduce the cohesiveness and improve the flow
 - Hygroscopic powders, stored and processed under low humidity conditions

Improvement...

4. Formulation additives (Flow activators)

- Flow activators are commonly referred as **glidants**
- Flow activators improve the flowability of powders by reducing adhesion and cohesion
- e.g. Talc, maize starch

Questions?

